

Conversational Graph Extraction

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Headline Claim

Seo et al.'s structural metrics for schema quality, and LLM-simulated users seem to only track ground-truth utility within narrow conditions. This project maps where they hold and where they break.

Research Questions and Scope

RQ1. Do Seo et al.'s six structural KG metrics predict downstream QA accuracy across schemas of varying quality? At least well enough to substitute for gold-ontology evaluation?

RQ2. Does feedback from an LLM-simulated user match feedback from human users, well enough to substitute simulated users for real users during development?

What is Interesting

- Seo's six metrics measure graph shape. Whether shape predicts utility is an empirical question.
- Simulated users may be biased — a yes-man, an over-merger, an under-pruner — and the bias is invisible if you never compare to humans. They could also have prior knowledge that the normal human does not.

Initial Thoughts on Experimentation

Corpora (two, for cross-domain check): still undecided on this.

RQ1 plan:

- Produce schema variants per corpus by varying dialogue
- Score each schema with all six Seo metrics + downstream QA accuracy.
- Measure within and across domain.

RQ2 plan:

- Get at least 5 humans (hopefully) to issue edits and then the simulated users to do the same, compare the responses
- Measure if schema's end up in the same place

Initial System Architecture

Component	Role
Corpus loader	Loads + samples docs
Schema store	JSON schema
Corruption module	Induces RQ1 schema-quality variants
Metric	Seo's six metrics + QA accuracy
Logger	JSON run records

Open Questions

- What counts as the "same domain" between two corpora?
- How to give a simulated user and a human equivalent access to the corpus, so the comparison is fair?
- How to score Seo metrics against any reference when no gold ontology exists?
- Is 300 docs per corpus enough to drive a measurable QA signal?

State of the Art — What is Relevant

- **Zhang & Soh (2024)** — **EDC**. Three-phase LLM pipeline (extract, define, canonicalize). Reusable as my extraction backbone.
- **Pan et al. (2025)** — **SCHEMA-MINER**. Human-in-the-loop schema mining with iterative LLM + expert feedback. Direct ancestor of the conversational setting.
- **Kommineni et al. (2024)**. Uses an LLM-as-judge for evaluation — precedent for simulated feedback.
- **Seo et al. (2022)**. Six ground-truth-free structural KG metrics. Direct subject of RQ1.
- **Wang et al. (2024)**. Survey of KG evaluation metrics. Maps the gaps RQ1 addresses.